# FLEET MOVEMENT, FUEL, DRIVER & MAINTENANCE MANAGEMENT SYSTEM

*Functional Requirements, Workflow & Screen Specifications*

**Phase 1: Mobile/Tablet-Based System Without GPS, RFID or OBD-II Dependency**

Purpose: Developer-ready functional specification for a centralized fleet control system covering vehicle movement, photographic odometer evidence, driver identification, fuel consumption, maintenance, tyres, document expiry, alerts, analytics and scheduled management reporting.

## 1. Project Objective

Develop a centralized Fleet Management System that initially operates without GPS trackers, RFID, or OBD-II. The primary operational device will be a mobile phone or tablet used by the gate/security person. Vehicle identity will be confirmed through a vehicle-specific QR/barcode and photographic evidence; driver identity will be captured from the company ID card; odometer readings will be photographed at Gate-Out and Gate-In.

The architecture must remain future-ready for GPS, OBD-II, CAN bus, RFID and fuel-sensor integrations without redesigning the core trip, vehicle, driver, fuel or maintenance database.

## 2. Core Operating Principle

The odometer is the central control reference. Every Gate-In/Out, fuel entry, oil change, tyre installation/replacement and mileage-based maintenance transaction must validate against the vehicle's odometer history.

Core lifecycle: Vehicle Registered → Driver Identified → Trip Requested/Authorized → Gate-Out → Vehicle/Odometer Photographed → Driver ID Captured → Trip Performed → Fuel Recorded → Gate-In → Closing Odometer Photographed → Trip KM Calculated → Fuel Efficiency Updated → Maintenance Mileage Updated → Alerts Checked → Reports/Emails Generated.

## 3. Main Modules

- Management Dashboard
- Vehicle Master
- Driver Master
- Vehicle Requisition / Authorization
- Vehicle Gate-Out
- Vehicle Gate-In
- Vehicles Currently Outside
- Trip Register

- Fuel Management
- Maintenance Management
- Tyre Management
- Vehicle Documents & Expiry
- Driver Licence & Document Expiry
- Alerts & Notifications
- Reports & Analytics
- Scheduled Email Reporting
- User Management & Permissions
- Audit Trail
- Administration / Configuration

## 4. User Roles

| Role | Main Rights |
|---|---|
| Administrator | Full configuration, masters, corrections, reports, users, alerts and audit access. Corrections require reason and remain auditable. |
| Fleet / Transport Manager | Review trips, fuel, maintenance, alerts, driver/vehicle performance and reports. |
| Gate Security Guard | Simple Gate-Out/Gate-In, QR scan, camera capture, driver ID capture, odometer entry, purpose/destination and remarks. |
| Management | Read-only executive dashboard and scheduled reports. |
| Driver (Optional Phase) | View assigned trip; upload fuel/emergency repair receipts and remarks. |

## 5. Vehicle Master

- Internal Vehicle ID (e.g., VEH-001)
- Registration number
- Company / business unit
- Make, model, variant, year and colour
- Fuel type
- Engine and chassis numbers
- Purchase/acquisition date and cost
- Department / cost centre
- Assigned driver, if applicable
- Expected fuel average (KM/L)
- Current validated odometer
- Vehicle status: Available / Outside / Workshop / Inactive
- Vehicle photograph
- Maintenance intervals
- Document details and expiry dates

Each vehicle must have a durable QR/barcode sticker at a defined location. The visible vehicle registration number should also be printed/pasted beside the code. The QR/barcode should contain an internal vehicle identifier, not sensitive vehicle data.

## 6. Driver Master

- Driver ID / Employee ID
- Name and photograph
- Company ID card image / QR or barcode if available
- CNIC and mobile
- Driving licence number, category, issue and expiry dates
- Company / department
- Assigned vehicle, if any
- Employment/status field
- Emergency contact and remarks

At Gate-Out the guard should scan the company ID QR/barcode where available; otherwise select the driver and capture the company ID card photograph.

## 7. Vehicle Requisition & Authorization

Where management wants prior authorization, a trip requisition should be created before Gate-Out. It should record requesting person, department, vehicle (optional until assignment), required date/time, purpose, destination, expected return and approver. The system may also permit authorized ad-hoc Gate-Out transactions where company policy allows.

- Configurable purposes: Bank Work, Customer Visit, Supplier Visit, Purchase, Government Department, FBR, Customs, PRA, SECP, Staff Transport, Factory/Warehouse Transfer, Courier/Documents, Maintenance/Workshop, Airport, Management Duty, Delivery, Collection, Other.
- Destination may be predefined or free text.
- Requested By and Department should support reporting of fleet usage by department.

## 8. Gate-Out Workflow

1. Guard selects VEHICLE OUT on the mobile/tablet.
2. Guard scans vehicle QR/barcode. System retrieves registration, make/model, company, fuel type, last odometer and current status.
3. System blocks Gate-Out if the vehicle is already outside.
4. Guard captures vehicle/QR identification photograph if required and mandatory odometer photograph through the application.
5. Guard manually enters Odometer OUT. System compares it with the latest validated odometer and rejects a lower reading unless an authorized override with reason is used.
6. Guard scans/selects the driver and captures the company ID card image where required. System checks licence status and warns/blocks according to configured policy.

7. Guard selects purpose, destination, requested by, department, approved by (if applicable), expected return and remarks.
8. Confirmation screen shows all key data and evidence before submission.
9. On confirmation, the system generates a unique Trip Number, records automatic server timestamp/user, and changes vehicle status to OUTSIDE.

## 9. Photograph & Evidence Controls

- Prefer direct camera capture inside the mobile/tablet interface; gallery upload should be restricted for critical Gate-Out/Gate-In evidence where technically feasible.
- Store original image plus transaction ID, vehicle ID, photo type, user and server timestamp.
- Photo types: OUT ODOMETER, IN ODOMETER, VEHICLE ID/QR, DRIVER ID, FUEL RECEIPT, MAINTENANCE INVOICE, DAMAGE/INCIDENT.
- System may create a display watermark: Vehicle No. | OUT/IN | Date | Time | Guard/User.
- Images must not be silently replaceable. Any authorized replacement/deletion must be logged in the Audit Trail.

## 10. Vehicles Currently Outside

| Vehicle | Driver | Out Time | Odo Out | Purpose | Destination | Duration / Status |
|---|---|---|---|---|---|---|
| LEA-1234 | Muhammad Aslam | 09:18 | 128,450 | Bank Work | Gulberg | 2h 15m / Normal |
| LEX-7865 | Imran | 10:05 | 83,412 | Purchase | Shah Alam | 1h 28m / Normal |

Duration should update automatically. Configurable status indicators may include Normal, Expected Back Soon and Overdue.

## 11. Gate-In Workflow

10. Guard selects VEHICLE IN and scans the vehicle QR/barcode.
11. System finds the vehicle's single open trip and displays Trip No., driver, Out time and Odometer OUT.
12. Guard captures a fresh IN odometer photograph through the application.
13. Guard enters Odometer IN. System rejects a value below Odometer OUT.
14. System automatically calculates Trip KM = Odometer IN − Odometer OUT and Trip Duration = Gate-In Time − Gate-Out Time.
15. Guard records return condition: Vehicle OK / Maintenance Required / Damage or Incident, plus remarks and optional issue categories.
16. Guard indicates whether fuel was purchased during the trip and whether maintenance attention is required.
17. On confirmation, trip becomes COMPLETED, vehicle becomes AVAILABLE, current odometer is updated, and maintenance/tyre mileage counters are recalculated.

## 12. Fuel Management

Fuel must be maintained as a separate but linked transaction because fuel may be filled before, during or after a trip, or without a specific trip.

| **Field** | **Requirement** |
|---|---|
| Vehicle | Scan QR or select |
| Date/Time | Automatic; editable only with permission |
| Odometer | Required and validated |
| Fuel Type | Petrol / Diesel / Other |
| Litres | Required |
| Rate/Litre | Required where cost tracking is used |
| Total | Auto-calculated |
| Fuel Station | Configured list / Other |
| Payment Method | Cash / Credit / Fuel Card / Company Account / Driver Paid / Other |
| Receipt No. | Optional/configurable |
| Driver | Linked |
| Full Tank? | Yes / No |
| Receipt Image | Mandatory where policy requires |

Preferred fuel efficiency method: Full-Tank-to-Full-Tank. The system should also calculate period averages and rolling averages. Vehicle benchmarks should allow expected-versus-actual fuel analysis and excess consumption alerts.

## 13. Fuel Analytics

- Trip/period KM
- Litres consumed
- KM/L
- Fuel cost
- Fuel cost per KM
- Expected KM/L benchmark
- Expected fuel for actual KM
- Actual fuel
- Excess/short fuel litres
- Estimated excess fuel cost
- Vehicle-wise and driver-wise averages
- Weekly, monthly and annual trends

Example: 2,500 KM at a 10.0 KM/L benchmark implies 250 L expected fuel. If actual fuel is 289 L, the system flags 39 L excess consumption for management review.

## 14. Maintenance Management

Maintenance must support mileage-based, date/time-based, or 'whichever comes first' schedules. Every maintenance transaction records vehicle, date/time, odometer, workshop/vendor, service items, parts/brands, quantities, labour, total cost, invoice/receipt image and remarks.

### 14.1 Maintenance Categories

| Category | Typical Items |
|---|---|
| Engine | Engine oil, oil filter, air filter, fuel filter, spark/glow plugs, tuning, injectors, timing belt/chain, coolant, radiator |
| Transmission | Gear oil, ATF, differential oil, clutch, transmission service |
| Brakes | Pads, shoes, brake oil, discs, brake service |
| Suspension | Shocks, bushes, ball joints, tie rods, alignment, balancing |
| Electrical | Battery, alternator, starter, lights, wiring |
| Air Conditioning | AC service, gas, compressor, cabin filter |
| Tyres | New tyre, rotation, alignment, balancing, puncture, replacement |
| Other | Wipers, belts, bearings, body/paint, accident repair, general service |

### 14.2 Maintenance Record Example

- Vehicle: LEA-1234
- Date: 22-Aug-2026
- Odometer: 130,150 KM
- Maintenance: Engine Oil Change
- Oil: Shell Helix HX7, 10W-40, 5.5 L
- Oil Filter: Changed; brand/part no./cost
- Air Filter / Fuel Filter / Plugs / Gear Oil / Tuning: Yes/No with details
- Labour and other parts cost
- Total maintenance cost automatically calculated
- Invoice/receipt image
- Next service interval: e.g., 5,000 KM and/or 6 months

If oil is changed at 130,150 KM with a 5,000 KM interval, next mileage due is 135,150 KM. If the time interval is six months, the due condition is whichever comes first.

## 15. Maintenance Alert Logic

| Status | Example Rule |
|---|---|
| Normal | More than 1,000 KM remaining |
| Due Soon | 500–1,000 KM remaining |
| Urgent | Less than 500 KM remaining |
| Overdue | Interval exceeded |

Alert thresholds must be configurable by maintenance item and vehicle type.

## 16. Tyre Management

- Unique Tyre ID
- Brand, model, size and serial number
- Purchase date and cost

- Installation date and odometer
- Vehicle and wheel position
- Expected mileage life
- Rotation history
- Removal date and odometer
- Removal reason
- Current status / spare / scrap / store

Tyre mileage = Current Vehicle Odometer − Installation Odometer, adjusted for periods when the tyre was removed or moved if detailed tyre-position tracking is enabled.

## 17. Vehicle & Driver Documents

- Vehicle: Registration/Smart Card, Token Tax, Insurance, Fitness Certificate, Route Permit, leasing documents, tracker subscription when applicable, other documents.
- Driver: Driving licence and other configured documents.
- Each document should support number, issue date, expiry date, attachment/image and alert lead time.
- Typical alerts: 30 days, 15 days, 7 days and Expired/Critical.

## 18. Management Dashboard

| KPI | Example |
|---|---|
| Total Vehicles | 24 |
| At Factory | 16 |
| Currently Outside | 6 |
| Workshop | 2 |
| KM Today | 864 KM |
| Fuel Today | 92 L |
| Fuel Cost Today | Rs. 27,450 |
| Trips Today | 31 |
| Average Trip | 27.9 KM |
| Maintenance Due | 3 vehicles |
| Tyre Alerts | 2 vehicles |
| Document Alerts | 1 vehicle |
| Fuel Exceptions | 2 vehicles |

Dashboard cards should drill down to the underlying vehicles/trips where practical.

## 19. Vehicle Profile / Analytics Screen

- Overview: current odometer, status, assigned driver, last trip, last fuel entry, last service.
- Period KPIs: Today / Week / Month / Year KM, trips, fuel, fuel cost, average KM/L.
- Cost analytics: maintenance cost, tyre cost, other vehicle cost, total operating cost and cost per KM.
- Maintenance: last/next oil change, filters, gear oil, tyres, battery and other schedules.
- Tabs: Overview, Trips, Fuel, Maintenance, Tyres, Documents, Costs, Photos, Audit History.

## 20. Driver Analytics

- Total trips and KM driven
- Vehicles driven
- Fuel average while driving each vehicle
- Average trip duration
- Fuel cost associated with trips
- Abnormal consumption cases
- Incidents / damage reports
- Late/overdue returns
- Licence/document status

## 21. Reports

| Report | Minimum Contents |
|---|---|
| Daily | Trips, vehicles outside, KM, fuel, unusual events, maintenance/actions required |
| Weekly | Vehicle, trips, KM, fuel litres/cost, KM/L, maintenance cost, total cost, cost/KM |
| Monthly | Opening/closing odometer, KM, trips, fuel, fuel cost, KM/L, maintenance, total direct cost, cost/KM |
| Annual | Opening/closing odometer, total KM/trips, fuel litres/cost, average KM/L, maintenance, tyres, insurance, other cost, total annual cost, cost/KM |
| Driver-wise | Trips, KM, vehicles, fuel performance, incidents and exceptions |
| Department-wise | Trips, KM, fuel cost and fleet utilization by requesting department |

## 22. Scheduled Email Reporting & Action Alerts

Administrator must configure designated recipients, report frequency and timing. Reports should be generated automatically and emailed at required intervals, subject to deployment environment and email service configuration.

- Daily fleet report: current outside vehicles, day trips/KM/fuel and exceptions.
- Weekly fleet summary.
- Monthly fleet performance report.
- Annual fleet report.
- Immediate/periodic action alerts: oil/service due, tyres approaching replacement, licence/insurance expiry, abnormal fuel average, overdue vehicle return.

Example action alert: 'LEA-1234 has approximately 450 KM remaining before scheduled engine oil replacement.'

## 23. Mobile / Tablet Gate Interface

The guard interface must be touch-first, fast and simple, with large controls and minimal typing.

| Main Action | Behaviour |
| --- | --- |
| VEHICLE OUT | Scan QR → capture odometer → identify driver → trip purpose/destination → confirm |
| VEHICLE IN | Scan QR → retrieve open trip → capture closing odometer → condition → close trip |
| CURRENTLY OUT | Live list of all open trips |
| FUEL ENTRY | Scan/select vehicle → odometer → fuel details → receipt image |

## 24. Suggested Screen: Vehicle Out

VEHICLE OUT

Vehicle: [SCAN QR]
LEA-1234 | Toyota Hilux

Odometer: [TAKE PHOTO]   [128450] KM

Driver: [SCAN/SELECT DRIVER]
Muhammad Aslam   [TAKE ID PHOTO]

Purpose: [Bank Work ▼]
Destination: [Gulberg Lahore]
Requested By: [Accounts ▼]
Expected Return: [Optional]
Remarks: [........................]

[CONFIRM VEHICLE OUT]

## 25. Suggested Screen: Vehicle In

VEHICLE IN

[SCAN VEHICLE QR]
Open Trip: TRP-2026-008541
Out Time: 09:18 AM
Odometer Out: 128,450 KM

Closing Odometer: [TAKE PHOTO]   [128578]
Calculated Trip KM: 128 KM
Return Time: Automatic

Condition: ○ OK  ○ Maintenance Required  ○ Damage/Incident
Remarks: [........................]

[COMPLETE GATE-IN]

## 26. Suggested Screen: Maintenance

VEHICLE MAINTENANCE

Vehicle: [LEA-1234 ▼]
Odometer: [130150]
Date/Time: Automatic
Workshop/Vendor: [................]

Service Items:
☐ Engine Oil ☐ Oil Filter ☐ Air Filter ☐ Fuel Filter
☐ Gear Oil ☐ Spark Plugs ☐ Tuning ☐ Brake Service
☐ Tyres ☐ Battery ☐ Other

Selecting an item opens its relevant brand, specification, quantity, cost and next-interval fields.

[SAVE MAINTENANCE]

## 27. Suggested Vehicle Dashboard

LEA-1234 | Toyota Hilux
Current Odometer: 134,650 KM | Status: AT FACTORY | Driver: Muhammad Aslam

THIS MONTH
KM: 2,580 | Trips: 52 | Fuel: 267 L | Average: 9.66 KM/L
Fuel Cost: Rs. 78,498 | Maintenance: Rs. 21,600

MAINTENANCE
Oil: 500 KM Remaining | Tyres: 7,800 KM Remaining | Insurance: 18 Days Remaining

## 28. Validation & Business Rules

18. Do not allow Gate-Out for a vehicle already outside.
19. Do not allow Gate-In where no open trip exists.
20. Do not allow closing odometer below opening odometer.
21. Do not allow a new odometer reading below the latest validated vehicle odometer without authorized override.
22. Do not allow duplicate active trips for the same vehicle.
23. Warn/block the same driver being assigned to two simultaneous active vehicles unless authorized.
24. Warn/block an expired driving licence according to configured policy.
25. Maintenance odometer materially below the current vehicle odometer requires authorized correction and reason.

26. Critical timestamps should use server/system time and ordinary users should not edit them.
27. Every administrative override must require a reason and be audit logged.

## 29. Audit Trail

Every critical action and correction must record user, date/time, transaction, previous value, new value and reason. Example: 'Odometer changed from 128,578 to 128,587 by Admin on 22-Aug-2026 at 17:43. Reason: incorrect manual entry verified from odometer photograph.'

## 30. Technical / Architecture Requirements

- Responsive web application / PWA recommended for Android mobile/tablet and desktop browser use.
- Central database with role-based access.
- Camera integration for direct evidence capture.
- QR/barcode scanning through device camera where supported.
- Secure image/document storage linked to transactions.
- Server-side validation for critical odometer and status rules.
- Configurable company/business unit structure.
- Configurable maintenance items, intervals, fuel benchmarks, purposes, departments and alert thresholds.
- Scheduled report generation and email service integration.
- Backups and restore procedures.
- Audit logs should not be editable by ordinary users.

## 31. Future Integration Fields

- GPS distance and coordinates
- Speed and route data
- Ignition status
- OBD/ECU odometer
- Engine hours
- Fuel level / fuel sensor data
- RPM
- Fault codes
- Geofence events
- RFID driver/vehicle identification

These future integrations must supplement, not invalidate, the original photographic/manual audit trail.

## 32. Recommended Development Phases

| Phase | Scope |
|---|---|
| Phase 1 | Vehicle/Driver Master, QR identification, requisition, Gate-Out/In, odometer/ID photos, trip register, basic dashboard |

| | |
|---|---|
| Phase 2 | Fuel management, fuel averages, benchmarks, exception alerts and reports |
| Phase 3 | Maintenance, oil/filters/gear oil/tuning/brakes/battery, tyres and mileage/date alerts |
| Phase 4 | Vehicle/driver documents, expiry alerts, scheduled email reporting |
| Phase 5 | Management analytics, cost/KM, driver/department analysis, weekly/monthly/annual reporting |
| Phase 6 – Future | GPS, OBD-II/CAN, RFID, fuel sensors, live tracking, speed monitoring and geofencing |

## 33. Developer Acceptance Summary

The finished Phase 1–5 system must provide an auditable digital replacement for manual vehicle gate, fuel and maintenance registers. Management should be able to answer, for any vehicle and any selected period: how many kilometres it ran; who drove it; why and where it went; when it left and returned; photographic odometer evidence; fuel consumed and average KM/L; fuel and maintenance cost; maintenance history; next required actions; tyre status; document expiries; and weekly/monthly/annual vehicle-, driver- and department-wise performance.